# ABSTRACT

Digital platforms have expanded rapidly, and competition among online services has intensified. As a result, understanding user behaviour has become critical for maintaining customer engagement over time. Customer churn remains a major challenge for e-commerce businesses. Users often reduce activity gradually or stop using a platform altogether. Traditional churn prediction systems mainly identify Total Churn, which delays intervention and overlooks users who show early signs of disengagement.

To overcome this limitation, this project proposes a multi-class churn prediction model that distinguishes between Non-Churn, Partial Churn, and Total Churn customers. The system processes multi-month e-commerce activity data to compute LRFM (Length, Recency, Frequency, Monetary) metrics along with behavioural indicators such as activity decline, usage normalization, purchase-to-session patterns, and spending changes. These features capture long-term customer value and recent behavioural shifts. Both are important.

The system applies K-Means clustering to group customers with similar behavioural patterns, improving interpretability and supporting retention planning. Multiple machine-learning models—including Logistic Regression, Decision Tree, Random Forest, SVM, Naive Bayes, MLP, and XGBoost—are trained using stratified datasets. A Hybrid Stacking Ensemble then combines their predictions to improve reliability. Model performance is evaluated using accuracy, precision, recall, F1-score, confusion matrices, and ROC curves, showing clear separation among churn categories.

By combining LRFM analysis, behavioural feature engineering, clustering, and ensemble learning, the proposed system provides a proactive approach to churn prediction. It helps identify early decline patterns and prioritize at-risk users. Action becomes possible. The architecture also supports future extensions such as time-series analysis, automated churn risk scoring, and real-time churn alerts.

# CHAPTER 01

# INTRODUCTION

## 1.1 DEFINITION

Customer churn describes the reduction or complete loss of user engagement with a digital platform over time. It directly affects business growth and stability, since retaining existing users usually costs less than acquiring new ones. In competitive digital markets, early churn identification has therefore become essential. Very essential.

In practice, churn does not occur in a single pattern and can be grouped into three categories:

- **Non-Churn:** Users who remain active with consistent engagement.
- **Partial Churn:** Users who continue using the platform but show reduced activity, interaction, or spending.
- **Total Churn:** Users who stop interacting with the platform entirely.

Among these categories, Partial Churn is the hardest to detect. Behavioural decline happens slowly. Often unnoticed. Identifying users at this stage allows businesses to act before complete disengagement occurs.

To handle this challenge, the proposed system adopts a three-class churn prediction framework that classifies users as Non-Churn, Partial Churn, or Total Churn. The system analyzes multi-period e-commerce data using LRFM (Length, Recency, Frequency, Monetary) metrics along with behavioural features such as activity-drop ratios, normalized usage patterns, and spending trends. K-Means clustering supports customer segmentation, while multiple machine-learning models are trained and combined using a Hybrid Stacking Ensemble to improve prediction stability. The system is evaluated using standard performance measures, enabling accurate detection of both early-stage and complete churn.

## 1.2 OBJECTIVES

The objective of this project is to design an effective churn prediction system that identifies Non-Churn, Partial Churn, and Total Churn users by analyzing customer behaviour across multiple time periods.

**The key objectives are:**

- To extract LRFM features from multi-month customer activity data in order to understand long-term engagement patterns.
- To develop behavioural indicators that highlight early signs of reduced user activity.
- To classify customers into three churn categories using a structured churn-labelling approach.
- To apply K-Means clustering for customer segmentation and behavioural pattern analysis.
- To train and compare multiple machine-learning models for churn prediction.
- To build a Hybrid Stacking Ensemble to improve prediction accuracy and stability.
- To evaluate system performance using accuracy, precision, recall, F1-score, confusion matrix, and ROC curves.

## 1.3 PROBLEM STATEMENT

Customer churn is a significant challenge for digital platforms, especially in e-commerce systems where user engagement changes over time. Most traditional churn prediction methods focus only on Total Churn, meaning customers who completely stop using the platform. This approach misses an important stage. Churn often develops gradually.

When systems fail to identify Partial Churn, businesses lose the opportunity to act early. Customers showing reduced activity remain unnoticed. In addition, many existing models rely on basic transactional or demographic data and ignore behavioural indicators such as activity decline, usage variation, and spending changes. As a result, prediction accuracy suffers.

There is therefore a need for an improved churn prediction framework that combines LRFM analysis, behavioural feature engineering, clustering, and machine-learning techniques. Such a system can identify multiple churn stages and support proactive customer management.

### 1.3.1 PROBLEM ANALYSIS

An examination of existing churn prediction approaches reveals several limitations:

- Most systems fail to detect Partial Churn, focusing only on active or inactive users.
- Traditional models depend on basic features and overlook detailed behavioural patterns.
- Binary churn classification does not represent the gradual nature of customer disengagement.

- Lack of clustering reduces interpretability of customer segments.
- Single-model approaches show inconsistent performance on complex behavioural data.
- Existing systems provide weak early-warning support for churn prevention.

To address these limitations, an effective churn prediction system must support multi-class classification, integrate behavioural and clustering features, and apply hybrid machine-learning techniques for reliable early detection.

## 1.4 PROPOSED SYSTEM

The proposed system presents a multi-stage churn prediction framework that identifies users across three churn categories: Non-Churn, Partial Churn, and Total Churn**.** It combines LRFM analysis, behavioural feature extraction, clustering, and machine-learning models to support early and accurate churn detection. Early action matters.

**Key components include:**

**Behavioural Feature Extraction:** The system computes LRFM metrics along with indicators such as activity decline, normalized usage, and spending trends. These features capture both long-term value and recent behaviour.

**Machine Learning and Hybrid Stacking:** Multiple classifiers are trained and their outputs are combined using a stacking ensemble. This improves prediction stability. Accuracy increases.

**Customer Segmentation and Evaluation:** The system applies K-Means clustering to group customers with similar behaviour and evaluates performance using standard metrics to verify effectiveness.

## 1.5 EXISTING SYSTEM

Most existing churn prediction systems use a binary classification approach that labels customers only as Churn or Non-Churn. These systems mainly depend on basic transactional data and do not account for gradual changes in user behaviour. Important signals are missed.

**Limitations of existing systems include:**

- Inability to detect Partial Churn, where users remain active but show declining engagement.

- Dependence on shallow features with little focus on behavioural patterns.
- Lack of multi-class classification and hybrid modelling, limiting prediction depth.
- Absence of customer segmentation, reducing interpretability and insight.

As a result, existing approaches generate mostly reactive outcomes and provide limited support for proactive and targeted retention strategies.

The scope of this project is to design a data-driven churn prediction system for e-commerce platforms with the following focus:

- Multi-class churn classification, covering Non-Churn, Partial Churn, and Total Churn customers.
- LRFM and behavioural feature engineering to capture long-term value and recent activity changes.

Customer segmentation using K-Means clustering to group users with similar behaviour patterns.

## 1.6 SCOPE OF THE PROJECT

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- Training and comparison of multiple machine-learning models to analyze different prediction strategies.
- Hybrid stacking ensemble implementation to improve prediction accuracy and consistency.
- System performance evaluation using standard metrics to validate prediction reliability.

# CHAPTER 2

# LITERATURE SURVEY

**Paper 1: Multi-Stage Churn Identification in Digital Commerce Platforms**

Patel and Srinivasan (2023) analyzed the shortcomings of binary churn prediction models, particularly their inability to capture early behavioural decline. They proposed a multi-stage churn approach using RFM features with Gradient Boosting classifiers to categorize customers as active, declining, or inactive. The framework improved interpretability and supported earlier churn detection. However, it did not analyze behavioural changes across multiple time periods. Gradual transitions remained unaddressed.

**Paper 2: Behavioural Feature Engineering for E-Commerce Churn Prediction**

Lohith and Menon (2022) focused on improving churn prediction by adding behavioural depth to transaction-based models. They introduced features such as session drop rate, spending variation, and activity trends, evaluated using Random Forest and SVM algorithms. The results showed noticeable accuracy gains over baseline methods. Still, churn was treated as a binary outcome. Partial churn was not identified.

**Paper 3: Customer Segmentation Using Clustering for Churn Management**

Das and Chauhan (2021) explored the use of customer segmentation to enhance churn prediction. They applied K-Means clustering to group users with similar behaviour and used Logistic Regression for churn classification. Segmentation improved interpretability and helped identify high-risk customer groups. However, clustering and churn prediction were handled as separate steps. Integration was missing.

**Paper 4: Machine Learning Approaches for Churn Prediction in Online Services**

Gupta and Roy (2020) compared several machine learning techniques, including Decision Tree, Naive Bayes, SVM, and Neural Networks, for churn prediction. Their analysis showed that ensemble-based approaches produced more stable results than individual models. The study did not explore hybrid stacking strategies or behavioural trend analysis. Important gaps remained.

**Paper 5: Predicting Customer Lifetime and Churn Using LRFM Analytics**

Huang and Lee (2019) examined customer value modelling using LRFM metrics for churn prediction. They applied Logistic Regression and Neural Networks to study long-term engagement patterns. The results showed improved churn prediction performance. Short-term behavioural variation was not considered. Partial churn detection was absent.

## COMPARISION ANALYSIS

| **Paper** | **Authors & Year** | **Problem Focus** | **Model & Approach** | **Key Aspects** | **Gap Identified** |
|---|---|---|---|---|---|
| Multi-Stage Churn Identification | Patel & Srinivasan (2023) | Binary churn limitation | RFM + Gradient Boosting | Early churn detection | No multi-period behaviour trends |
| Behavioural Feature Engineering | Lohith & Menon (2022) | Limited feature depth | Behavioural features + RF + SVM | Improved accuracy | No partial churn classification |
| Customer Segmentation | Das & Chauhan (2021) | No customer grouping | K-Means + Logistic Regression | Better interpretability | Not an integrated pipeline |
| ML Approaches for Churn | Gupta & Roy (2020) | Model instability | DT, NB, SVM, ANN | Ensemble stability | No hybrid stacking |
| LRFM-Based Churn | Huang & Lee (2019) | Lack of value modelling | LRFM + LR + ANN | Long-term engagement analysis | No behavioural features |

# CHAPTER 3

# SYSTEM REQUIREMENT SPECIFICATIONS

This chapter outlines the functional, non-functional, hardware, and software requirements of the proposed multi-stage customer churn prediction system. These requirements define how the system handles data, performs churn classification, and maintains dependable operation. Clarity matters.

## 3.1 FUNCTIONAL REQUIREMENTS

The functional requirements describe the core operations that the Customer Churn Prediction System must support to classify customers into Non-Churn, Partial Churn, and Total Churn categories. These functions drive system behaviour.

**Primary Functional Requirements**

- **Multi-Month Data Processing:** The system loads and merges customer activity data from multiple months, including October, November, and the Telco dataset. It handles missing values, standardizes column names, and combines all records into a single structured dataset. Clean data matters.
- **LRFM Feature Extraction:** The system computes Length, Recency, Frequency, and Monetary (LRFM) metrics for each customer to capture long-term engagement patterns relevant to churn analysis.
- **Behavioural Feature Engineering:** The system generates short-term behavioural indicators such as activity drop ratio, normalized frequency and monetary values, recency weighting, and purchase-to-session ratio. These features help detect Partial Churn early. Timing is critical.
- **Churn Label Generation:** The system classifies customers as Non-Churn, Partial Churn, or Total Churn based on changes in activity and purchasing behaviour across consecutive months.
- **Customer Segmentation:** The system applies K-Means clustering on LRFM features to group customers with similar behavioural characteristics. The assigned cluster ID is used as an additional feature during model training.
- **Machine Learning Model Training and Comparison:** The system trains and evaluates multiple machine-learning models, including Logistic Regression, Decision

Tree, Random Forest, SVM, Naive Bayes, MLP, and XGBoost. It evaluates each model using accuracy, precision, recall, and F1-score.

- **Hybrid Stacking Ensemble:** The system combines predictions from base models using a stacking approach, with Logistic Regression acting as the meta-learner to improve stability and multi-class prediction performance.
- **Model Evaluation and Visualization:** The system generates evaluation outputs such as accuracy comparison charts, precision–recall–F1 graphs, confusion matrices, and ROC curves. Results are clear.
- **Final Churn Prediction Output:** The system outputs final churn labels and highlights high-risk customers to support informed business decisions.

**Secondary Functional Requirements**

- **Model Persistence:** The system saves trained models as serialized .pkl files and reloads them when required for reuse or evaluation. Simple reuse.
- **Experiment Logging:** The system records logs related to dataset loading, feature generation, model training, and ensemble outputs. This helps tracking.
- **Error Handling:** The system displays clear error messages for missing files, invalid inputs, incorrect column names, and model loading issues. Errors stay visible.

## 3.2 NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements define the quality, reliability, and scalability of the churn prediction system. They guide system performance. They matter.

**Performance Requirements**

The system shall deliver accurate predictions, with strong focus on Partial Churn and Total Churn detection using ensemble models.

- Feature extraction and preprocessing shall execute efficiently to support large datasets.
- Model training shall remain optimized to avoid unnecessary computational load. Speed matters.

**Reliability Requirements**

The system shall produce consistent results when run multiple times on the same dataset.

- The ensemble framework shall minimize misclassification and improve prediction stability.
- The system shall maintain data integrity during preprocessing and feature computation.No silent errors.

**Scalability Requirements**

The system architecture shall allow future extensions such as real-time churn prediction and time-series analysis.

- The system shall process large customer datasets without noticeable performance loss. Growth ready.

**Security Requirements**

The system shall store model files and processed datasets securely to prevent unauthorized changes.

- The system shall handle customer data confidentially within controlled storage locations and Privacy first.

## 3.3 HARDWARE REQUIREMENTS

The system is designed to run on commonly available computing environments without the need for specialized hardware, Simple setup and Reliable execution.

**Minimum Configuration**

- **Processor:** Standard processor capable of running Python-based applications
- **Memory:** Minimum of 4 GB RAM
- **Storage:** 10–20 GB of available disk space
- **System Type:** Laptop or desktop system

**Recommended Configuration**

- **Processor:** Multi-core processor for faster execution
- **Memory:** 8–16 GB RAM for improved performance
- **Storage:** SSD storage (256 GB or higher)
- **GPU:** Not required; may be used for future deep learning extensions

The system supports multiple platforms, including Windows, Linux, macOS, and cloud-based virtual machines and platform flexibility matters.

## 3.4 SOFTWARE REQUIREMENTS

**Operating System**

- Windows 10 or 11
- Ubuntu 20.04 or later
- macOS (latest versions)

**Development Environment**

- **Programming Language:** Python 3.8 or higher
- **Libraries:** pandas, numpy, scikit-learn, xgboost, matplotlib, seaborn, joblib
- **IDE:** Visual Studio Code

**Storage and Data Management**

- Local directory structure for storing processed datasets, trained models, and evaluation results
- Optional database support (SQLite or MySQL) for future logging and analytics

# CHAPTER 4

# DATASET DESCRIPTION AND PREPROCESSING

The performance of a churn prediction system depends largely on data quality and the preprocessing applied before model training. Poor data leads to weak predictions. This project uses multi-month e-commerce activity datasets to track changes in customer behaviour over time. These datasets provide the transactional and behavioural details required for LRFM calculation, behavioural feature creation, and churn labelling.

## 4.1 DATASET DESCRIPTION

The system uses customer activity data collected across consecutive months to analyze behavioural variation. A separate benchmark dataset is also included to validate model generalization where validation matters.

### 4.1.1 October E-Commerce Dataset

The October dataset represents baseline customer behaviour and contains activity logs related to engagement and purchase actions. It is used to:

- Establish normal customer engagement patterns
- Compute Frequency and Monetary metrics
- Act as a reference point for behavioural comparison

This dataset reflects customer behaviour before churn-related changes begin.

### 4.1.2 November E-Commerce Dataset

The November dataset records customer activity in the following month using the same attributes as October. It is used to:

- Compare customer behaviour with the baseline month
- Detect reductions in engagement or spending
- Identify Partial and Total Churn patterns

This dataset plays a key role in observing behavioural decline over time.

### 4.1.3 Combined Dataset Description

After initial cleaning, the system merges the October and November datasets into a single dataset while retaining time-based information through timestamps where the temporal order matters.

The combined dataset is used to:

- Extract LRFM features across multiple months
- Analyze changes in customer behaviour over time
- Generate churn labels for classification
- Support customer clustering and model training

Although the data is merged for processing efficiency, the system maintains month-wise separation during feature computation.

| user_id<br>Permanent user ID | user_session<br>User session ID |
|---|---|
| 10.3m – 580m | 13776051 unique values |
| 520088904 | 4d3b30da-a5e4-49df-b1a8-ba5943f1dd33 |
| 530496790 | 8e5f4f83-366c-4f70-860e-ca7417414283 |
| 561587266 | 755422e7-9040-477b-9bd2-6a6e8fd97387 |
| 518085591 | 3bfb58cd-7892-48cc-8020-2f17e6de6e7f |
| 558856683 | 313628f1-68b8-460d-84f6-cec7a8796ef2 |

| event_time<br>When event is was happened (UTC) | event_type<br>Event type: one of [view, cart, remove_from_cart, purchase] | product_id<br>Product ID | category_id<br>Product category ID | category_code<br>Category meaningful name (if present) | brand<br>Brand name in lower case (if present) | price<br>Product price |
|---|---|---|---|---|---|---|
| 67501979 total values | view 94%<br>cart 4%<br>Other (916939) 1% | 1.00m – 100m | 2053013552 b – 2187707861 b | [null] 32%<br>electronics.smart... 24%<br>Other (29228808) 43% | [null] 14%<br>samsung 12%<br>Other (50394499) 75% | 0 |
| 2019-11-01 00:00:00 UTC | view | 1003461 | 2053013555631882655 | electronics.smartphone | xiaomi | 489.07 |
| 2019-11-01 00:00:00 UTC | view | 5000088 | 2053013566100866035 | appliances.sewing_machine | janome | 293.65 |
| 2019-11-01 00:00:01 UTC | view | 17302664 | 2053013553853497655 | | creed | 28.31 |
| 2019-11-01 00:00:01 UTC | view | 3601530 | 2053013563810775923 | appliances.kitchen.washer | lg | 712.87 |
| 2019-11-01 00:00:01 UTC | view | 1004775 | 2053013555631882655 | electronics.smartphone | xiaomi | 183.27 |
| 2019-11-01 00:00:01 UTC | view | 1306894 | 2053013558920217191 | computers.notebook | hp | 360.09 |
| 2019-11-01 00:00:01 | view | 1306421 | 2053013558920217191 | computers.notebook | hp | 514.56 |

**Figure 4.1.3:** Combined Datasets

### 4.1.4 Telco Customer Churn Dataset

The Telco dataset serves as a benchmark for validating machine-learning model performance. It includes predefined churn labels along with customer and service-related attributes. Used for comparison.

This dataset supports:

- Model validation to assess predictive reliability
- Cross-domain analysis to evaluate generalization across industries
- Performance benchmarking against established churn datasets

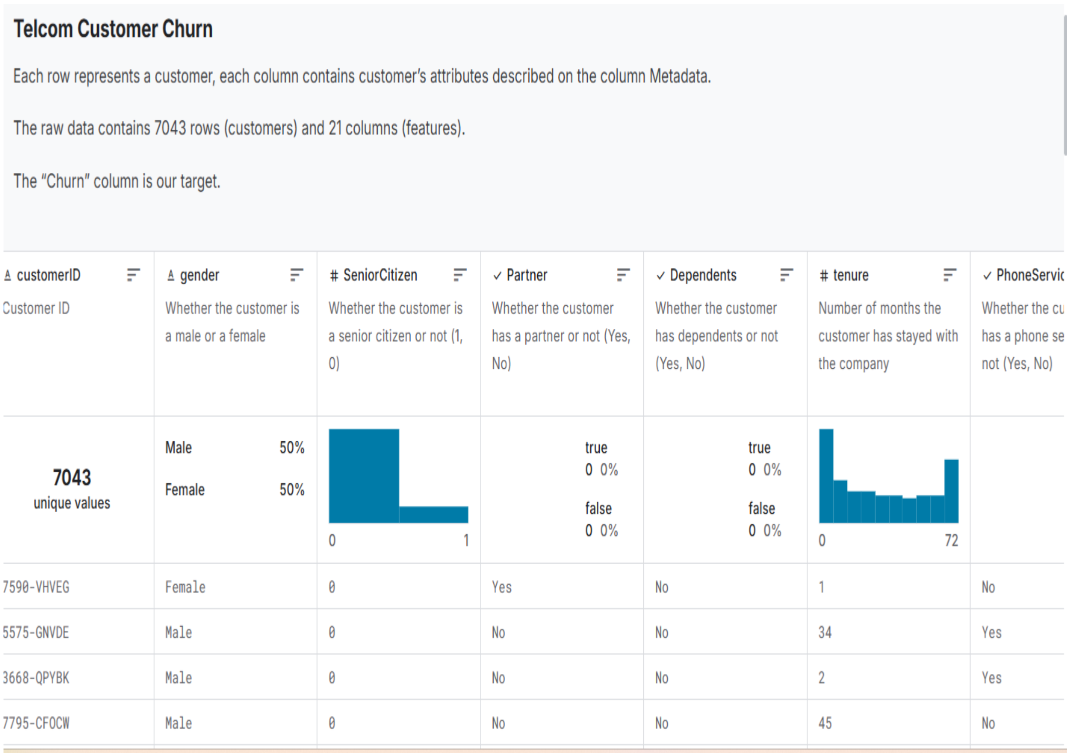


**Figure 4.1.4:**Telco Datasets

### Dataset Attributes

Common attributes across the datasets include:

- User identifier to uniquely represent each customer.
- Interaction or purchase frequency to measure engagement levels.
- Monetary transaction value to capture spending behaviour.
- Last activity timestamp to track recent user interaction.
- Account duration to represent the length of platform usage.

These attributes serve as the foundation for LRFM feature extraction.

## 4.2 DATA PREPROCESSING METHODOLOGY

Raw datasets often contain missing values, noise, and inconsistencies that affect analysis quality. A structured preprocessing pipeline is applied to prepare the data for reliable modelling. Clean data matters.

### 4.2.1 Data Cleaning

- Removal of missing or null values from critical columns
- Identification and removal of duplicate records
- Correction or exclusion of invalid data entries

This step ensures consistency.

### 4.2.2 Data Integration and Merging

The October and November datasets are merged using a unique user identifier to support direct month-to-month behavioural comparison. This comparison is essential for churn detection.

### 4.2.3 Data Type Standardization

- Numerical attributes are converted to appropriate numeric formats
- Date-related fields are transformed into datetime format
- Categorical values are standardized for uniform representation

This improves accuracy.

## 4.3 LRFM FEATURE COMPUTATION

After preprocessing, the following LRFM metrics are calculated:

- **Length (L):** Duration of the customer's association with the platform
- **Recency (R):** Time elapsed since the last recorded activity
- **Frequency (F):** Number of interactions or transactions
- **Monetary (M):** Total spending value

These metrics represent long-term behaviour.

## 4.4 BEHAVIOURAL FEATURE ENGINEERING

Additional behavioural indicators are generated to detect early churn signals:

- Activity drop ratio
- Normalized frequency and monetary values
- Recency weighting
- Purchase-to-session ratio
- Month-wise monetary trend

These features strengthen Partial Churn detection.

## 4.5 CHURN LABEL GENERATION

Based on LRFM patterns and behavioural changes, customers are categorized into:

- **Non-Churn:** Stable or increased engagement
- **Partial Churn:** Reduced but continued engagement
- **Total Churn:** Complete inactivity

This multi-class approach reflects real usage patterns.

## 4.6 FINAL DATASET PREPARATION

After preprocessing and feature generation:

- The cleaned dataset is stored in the processed directory
- Engineered features and cluster labels are appended
- The final dataset is prepared for model training and evaluation

Everything is ready.

## SUMMARY

This phase converts raw customer activity logs into a structured and reliable dataset. By combining multi-month data, LRFM metrics, behavioural features, and churn labels, the system builds a strong base for accurate multi-class churn prediction.

# CHAPTER 5

# MODEL DEVELOPMENT

This chapter describes the system architecture, module organization, algorithm design, and data flow of the proposed multi-stage customer churn prediction system. The design focuses on accurate identification of Non-Churn, Partial Churn, and Total Churn customers by combining behavioural analysis, clustering, and ensemble-based learning. Structure matters.

## 5.1 ARCHITECTURE SELECTION AND JUSTIFICATION

The architecture aims to support early churn detection while maintaining accuracy, clarity, and scalability. Instead of using opaque deep learning models, the system adopts a feature-driven machine learning approach. This choice improves interpretability.

**The main architectural components of the system are described below:**

- **LRFM-Based Behavioural Architecture:** LRFM metrics serve as the foundation of the system by representing long-term customer engagement and value patterns required for churn analysis.
- **Behavioural Feature Engineering Layer:** Short-term behavioural indicators such as activity drop, usage normalization, spending variation, and recency weighting are generated to detect early signs of Partial Churn. Small changes matter.
- **Clustering-Enhanced Architecture:** The system applies K-Means clustering to group customers with similar behaviour, improving interpretability and enabling segment-level churn analysis.
- **Hybrid Machine Learning Architecture:** Multiple machine learning models are trained in parallel, and their outputs are combined using a stacking-based ensemble to improve prediction stability and accuracy.
- **Modular and Scalable Design:** Each system component functions independently, allowing future extensions such as real-time churn prediction and time-series analysis. Growth is supported.

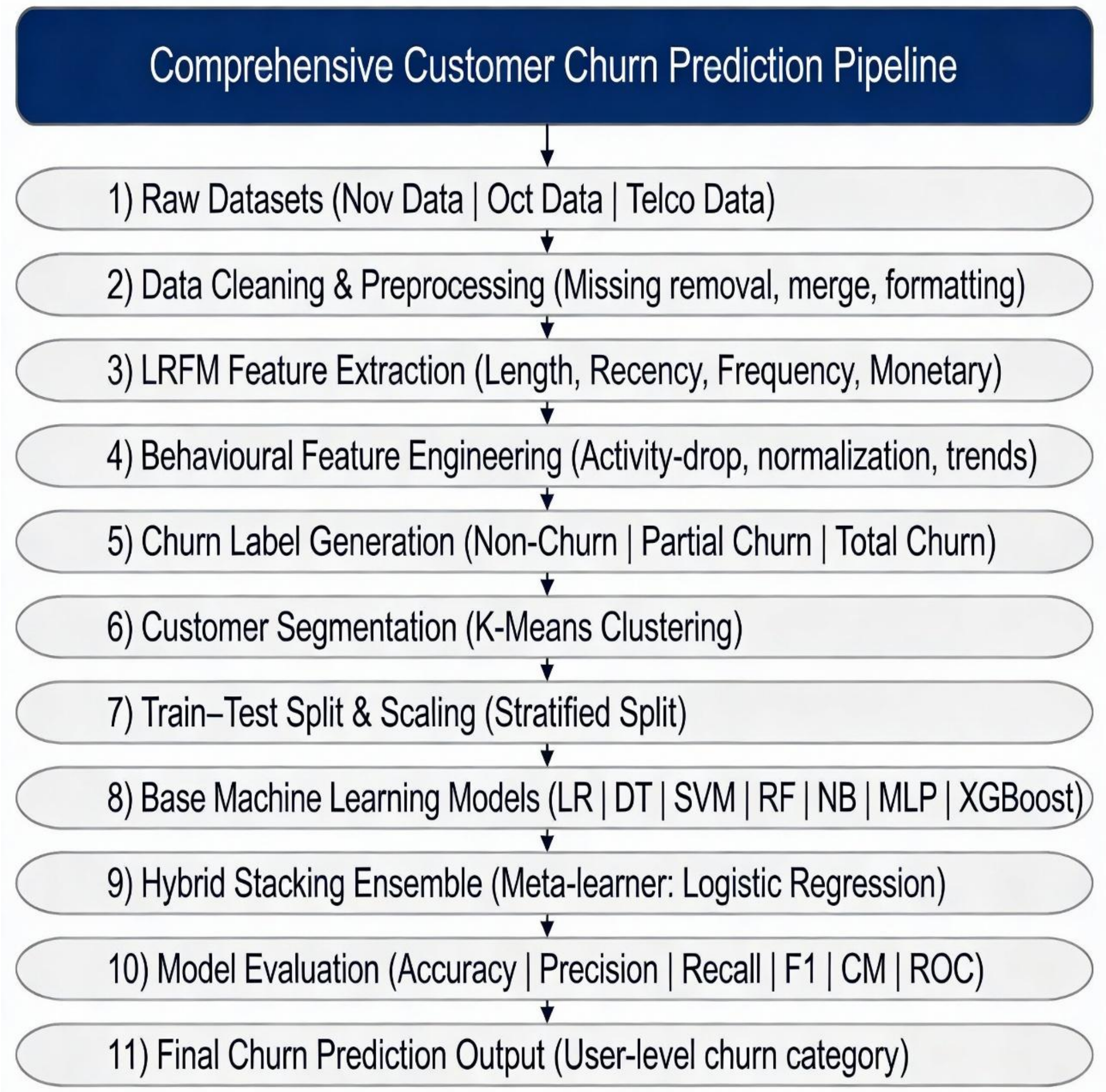


**Figure 5.1:** Architecture

## 5.2 MODULE ARCHITECTURE DETAILS

The system consists of multiple functional modules, each responsible for converting raw customer data into churn predictions. Every module has a clear role. No overlap.

### 5.2.1 Data Ingestion and Preprocessing Module

- Loads multi-month customer datasets
- Removes missing and inconsistent values
- Standardizes data formats
- Merges datasets using customer identifiers

This module ensures data quality and stability.

### 5.2.2 LRFM Feature Extraction Module

- Computes Length, Recency, Frequency, and Monetary metrics

This module captures long-term engagement patterns and separates loyal customers from inactive users.

### 5.2.3 Behavioural Feature Engineering Module

- Calculates activity-drop ratio
- Normalizes frequency and monetary values
- Computes purchase-to-session ratio
- Applies recency weighting and monetary trend analysis

These features support early detection of Partial Churn.

### 5.2.4 Churn Labelling Module

Customers are classified into:

- **Non-Churn:** Stable engagement
- **Partial Churn:** Reduced engagement
- **Total Churn:** Complete inactivity

This module converts behavioural changes into supervised learning labels.

### 5.2.5 Customer Segmentation Module (K-Means)

- Applies clustering on LRFM and behavioural features
- Assigns cluster identifiers to customers

This improves interpretability and churn risk grouping.

### 5.2.6 Machine Learning Model Training Module

The system trains multiple models:

- Logistic Regression
- Decision Tree
- Random Forest
- SVM
- Naive Bayes

- MLP
- XGBoost

This supports comparative evaluation and diverse pattern learning.

### 5.2.7 Hybrid Stacking Ensemble Module

- Combines probability outputs from base models
- Uses Logistic Regression as the meta-learner
- Generates final churn classification

This improves prediction stability and reduces classification errors.

### 5.2.8 Evaluation and Visualization Module

- Computes accuracy, precision, recall, and F1-score
- Generates confusion matrices and ROC curves

**Importance:**

- Confirms model reliability
- Ensures correct classification across all churn categories

## 5.3 UML DIAGRAMS

### 5.3.1 Use Case Diagram

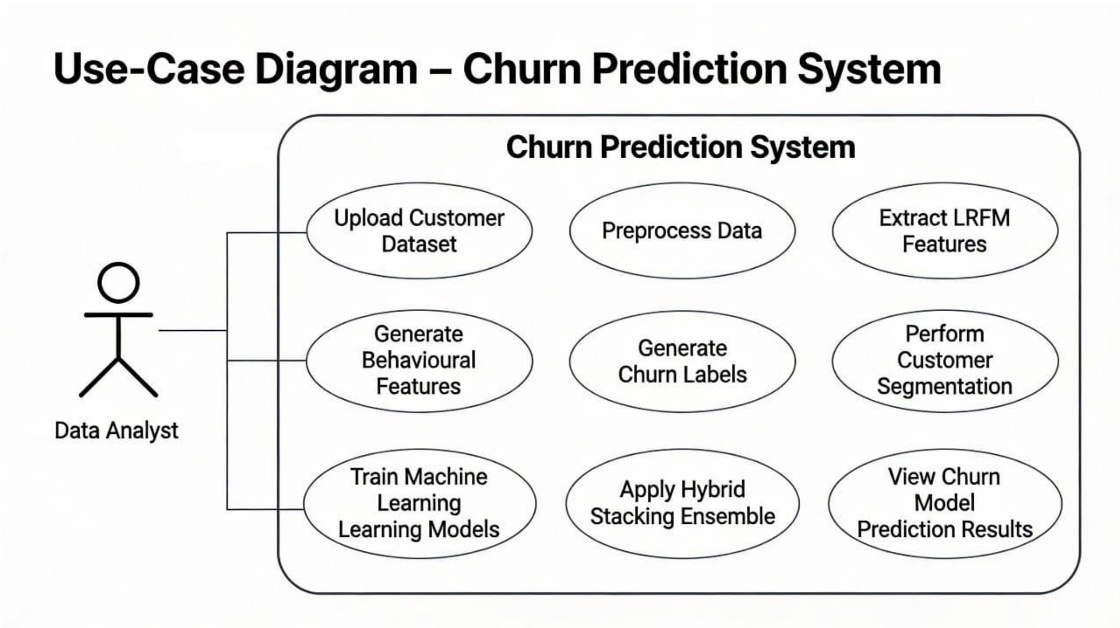


**Figure 5.3.1:** Use Case diagram

**Explanation:** The Use Case Diagram shows the functional interaction between the user and the Customer Churn Prediction System. It presents a high-level view of system operations and identifies who performs each action. Simple and clear.

**Actor**

**Data Analyst:** The Data Analyst acts as the primary user of the system. The analyst uploads customer datasets, runs churn analysis, trains models, and reviews prediction outputs. Decisions follow.

**Use Cases Description**

- **Upload Customer Dataset:** The Data Analyst uploads historical customer activity and transaction data, including October, November, and benchmark datasets.
- **Preprocess Data:** The system cleans the uploaded data by removing missing values, handling duplicate entries, standardizing formats, and merging datasets for analysis.
- **Extract LRFM Features:** The system computes Length, Recency, Frequency, and Monetary metrics to represent long-term customer engagement and value.
- **Generate Behavioural Features:** The system derives short-term behavioural indicators such as activity decline, usage trends, and spending changes to detect early disengagement.
- **Generate Churn Labels:** Based on month-wise behavioural variation, the system classifies customers into Non-Churn, Partial Churn, or Total Churn categories.
- **Perform Customer Segmentation:** The system groups customers into behaviourally similar clusters using K-Means clustering to support segment-level analysis.
- **Train Machine Learning Models:** The system trains multiple machine-learning models using the extracted features to predict churn behaviour.
- **Apply Hybrid Stacking Ensemble:** The system combines predictions from individual models using a stacking-based ensemble to improve accuracy and stability.
- **View Churn Prediction Results:** The Data Analyst reviews final churn predictions along with evaluation metrics for business analysis and decision-making.

**Summary:** The Use Case Diagram outlines the complete functional scope of the churn prediction system. It shows how the Data Analyst interacts with the system from data upload to final churn prediction, covering preprocessing, feature extraction, clustering, model training, and ensemble-based classification.

### 5.3.2 Activity Diagram

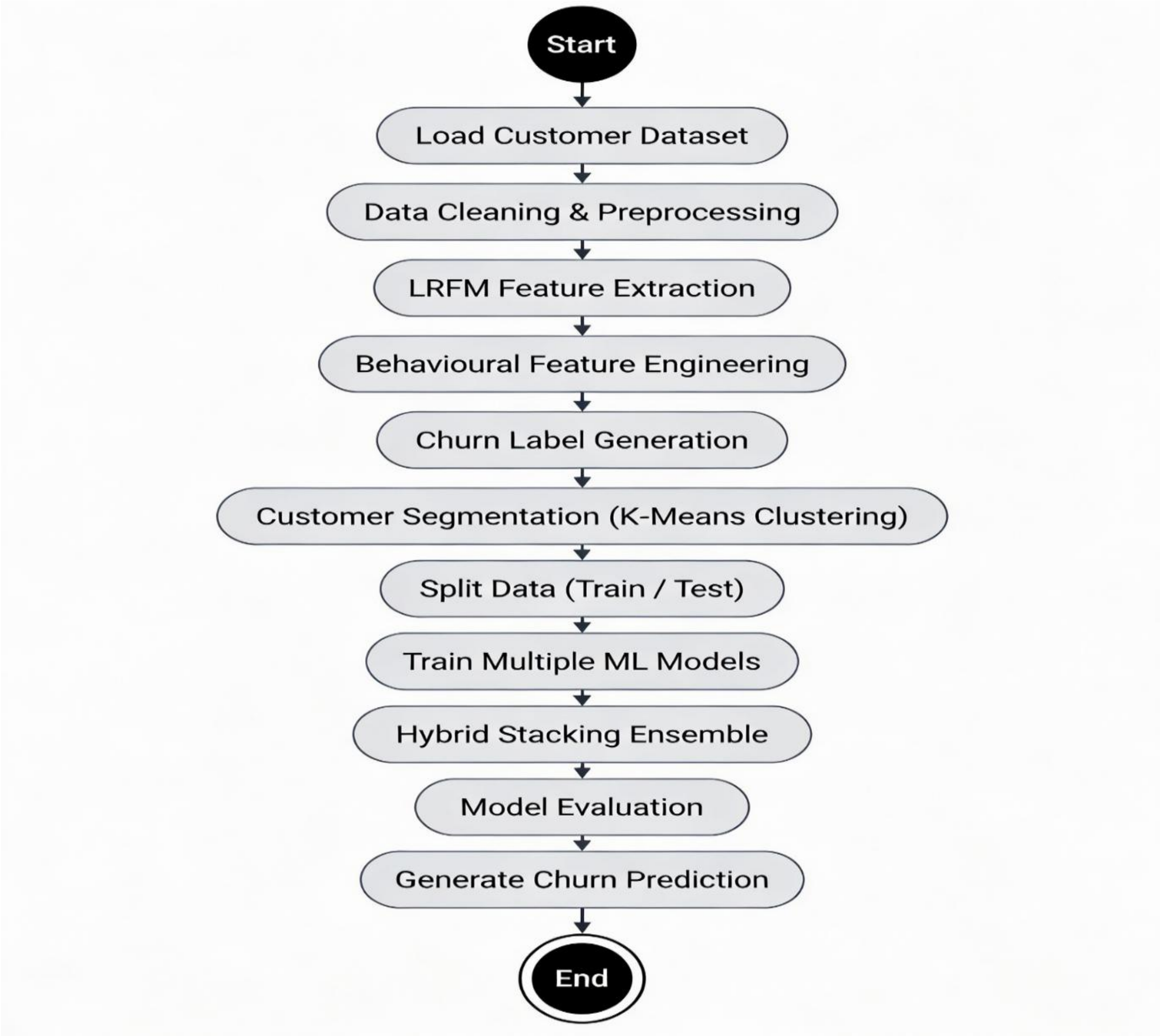


**Figure 5.3.2 :** Activity Diagram

**Explanation:** The Activity Diagram represents the complete workflow of the customer churn prediction process. It illustrates how the system moves from data input to final churn prediction. Step by step.

**Workflow Description**

The process starts when the user uploads customer activity datasets into the system.

- The system cleans and preprocesses the data to remove missing values and inconsistencies.
- LRFM feature extraction analyzes long-term customer behaviour.
- Behavioural feature engineering captures short-term engagement changes.

- K-Means clustering groups customers into similar behavioural segments.
- The system forwards the processed dataset to multiple machine learning models for training.
- Each model produces churn predictions independently.
- The hybrid stacking ensemble merges predictions from all models.
- The system evaluates performance using accuracy, precision, recall, and F1-score.
- The final churn category—Non-Churn, Partial Churn, or Total Churn—is generated.

**Summary:** The Activity Diagram presents a clear, flow-oriented view of system execution. It shows how customer data moves through preprocessing, feature extraction, modelling, evaluation, and final churn prediction stages.

### 5.3.3 Sequence Diagram

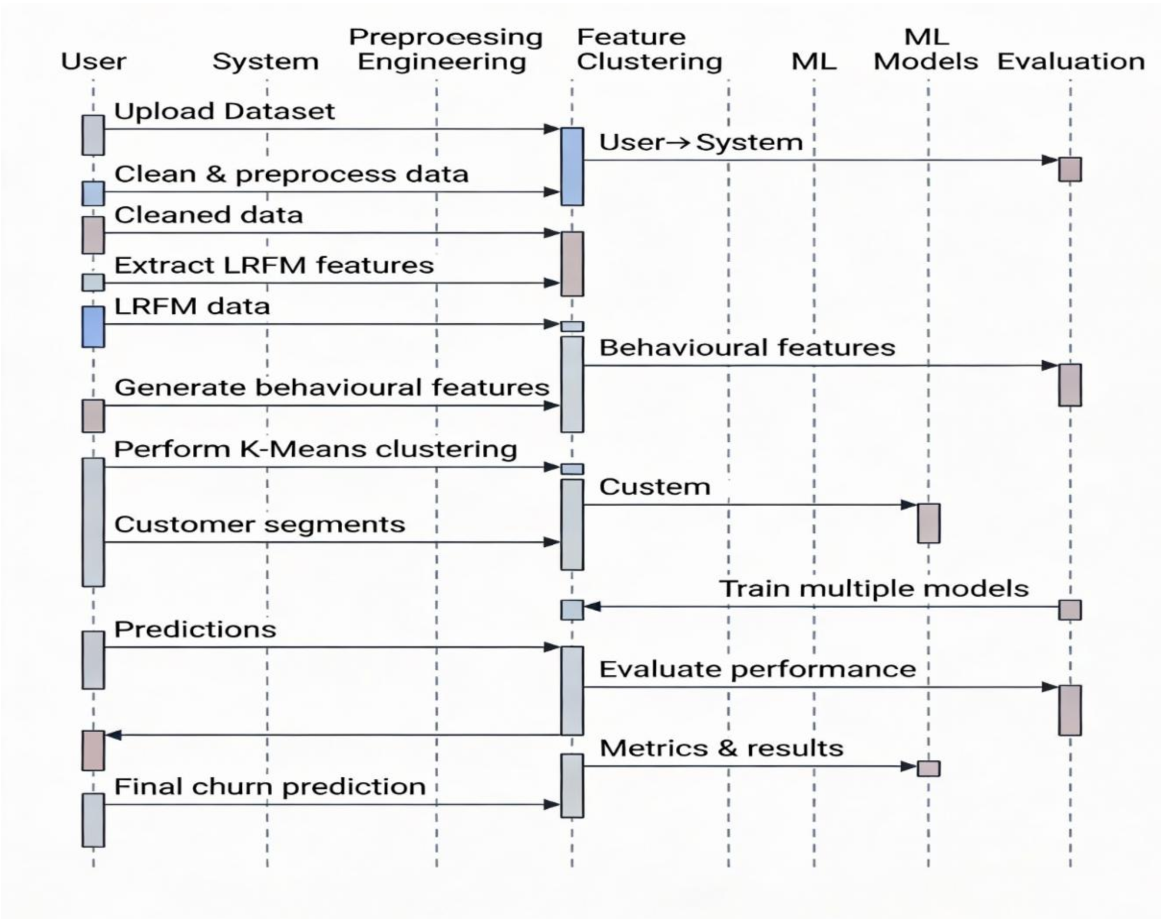


**Figure 5.3.3:** Sequence Diagram

**Explanation**: The Sequence Diagram represents the ordered interaction between system components involved in churn prediction. It shows how data moves step by step across modules and how each component exchanges information during execution. One step at a time.

**Interaction Flow**

The user starts the process by uploading the dataset.

- The preprocessing module cleans the data and applies formatting rules.
- The feature extraction module calculates LRFM metrics.
- The behavioural feature module generates short-term activity indicators.
- The churn labelling module assigns appropriate churn categories.
- The clustering module groups customers using K-Means.
- The training module forwards data to multiple machine learning models.
- Each model returns probability-based predictions.
- The ensemble module combines these predictions using a meta-learner.
- The evaluation module computes performance metrics.
- The system delivers final churn predictions to the user.

**Summary:** The Sequence Diagram outlines the execution order and coordination among system modules, ensuring that each operation follows a clear and logical flow.

## 5.4 ALGORITHM DEVELOPMENT AND OPTIMIZATION

This section describes the steps taken to improve algorithm efficiency and prediction accuracy. The focus remains on performance.

### 5.4.1 LRFM Computation Algorithm

**Optimizations applied:**

- Date-based recency values are calculated efficiently
- Frequency is computed using aggregated interaction counts
- Monetary values are normalized to maintain scale consistency

These steps support fast execution, even with large datasets.

### 5.4.2 Behavioural Feature Algorithm

**Optimizations applied:**

Vectorized operations are implemented using pandas

- Feature values are normalized to reduce scale bias
- Zero or missing frequency values are handled carefully

These changes improve feature reliability. Learning becomes stable.

**5.4.3 Clustering Algorithm (K-Means)**

**Optimizations applied:**

The elbow method is used to select an appropriate number of clusters

- Input features are standardized before clustering
- Iterations are reduced to speed up convergence

This ensures effective segmentation.

**5.4.4 Machine Learning Training Algorithm**

**Enhancements added:**

A stratified train–test split maintains class balance

- Feature scaling improves model consistency
- Stable hyperparameter settings reduce variance

Learning remains unbiased.

**5.4.5 Ensemble Optimization**

**Optimizations applied**

Probability-based stacking combines model outputs

- A lightweight meta-learner reduces complexity
- Overfitting is minimized through controlled model integration

Generalization improves.

## 5.5 MODEL ARCHITECTURE DETAILS

The system operates through three main architectural phases:

- **Customer Behaviour Analysis Phase:** Raw customer activity data is processed to generate LRFM and behavioural features. Each user receives a structured behaviour profile.
- **Prediction and Segmentation Phase:** Clustering and machine-learning models analyze behaviour profiles and produce churn predictions.
- **Decision and Output Phase:** The system assigns final churn labels—Non-Churn, Partial Churn, or Total Churn—along with evaluation results and insights.

## 5.6 DATA FLOW DIAGRAMS

### 5.6.1 Level 0 DFD – Churn Prediction System

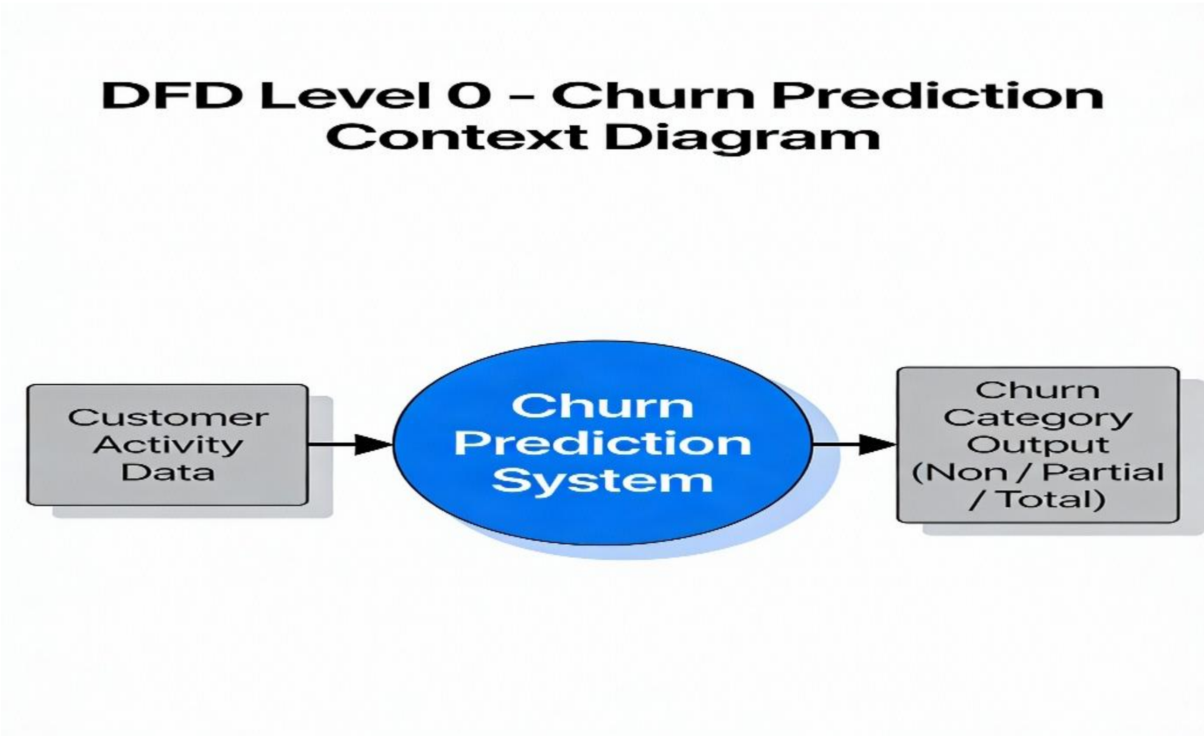


**Figure 5.6.1:** Level 0 DFD – Churn Prediction System

**Explanation:** In the Level 0 Data Flow Diagram, Customer Activity Data serves as the external input to the Churn Prediction System. The system internally analyzes interaction and transaction records using feature extraction and machine learning methods. The final output is the Churn Category, classifying customers as Non-Churn, Partial Churn, or Total Churn. This diagram presents a high-level system overview. Internal details remain hidden.

### 5.6.2 Level 1 DFD – Detailed Churn Prediction Process

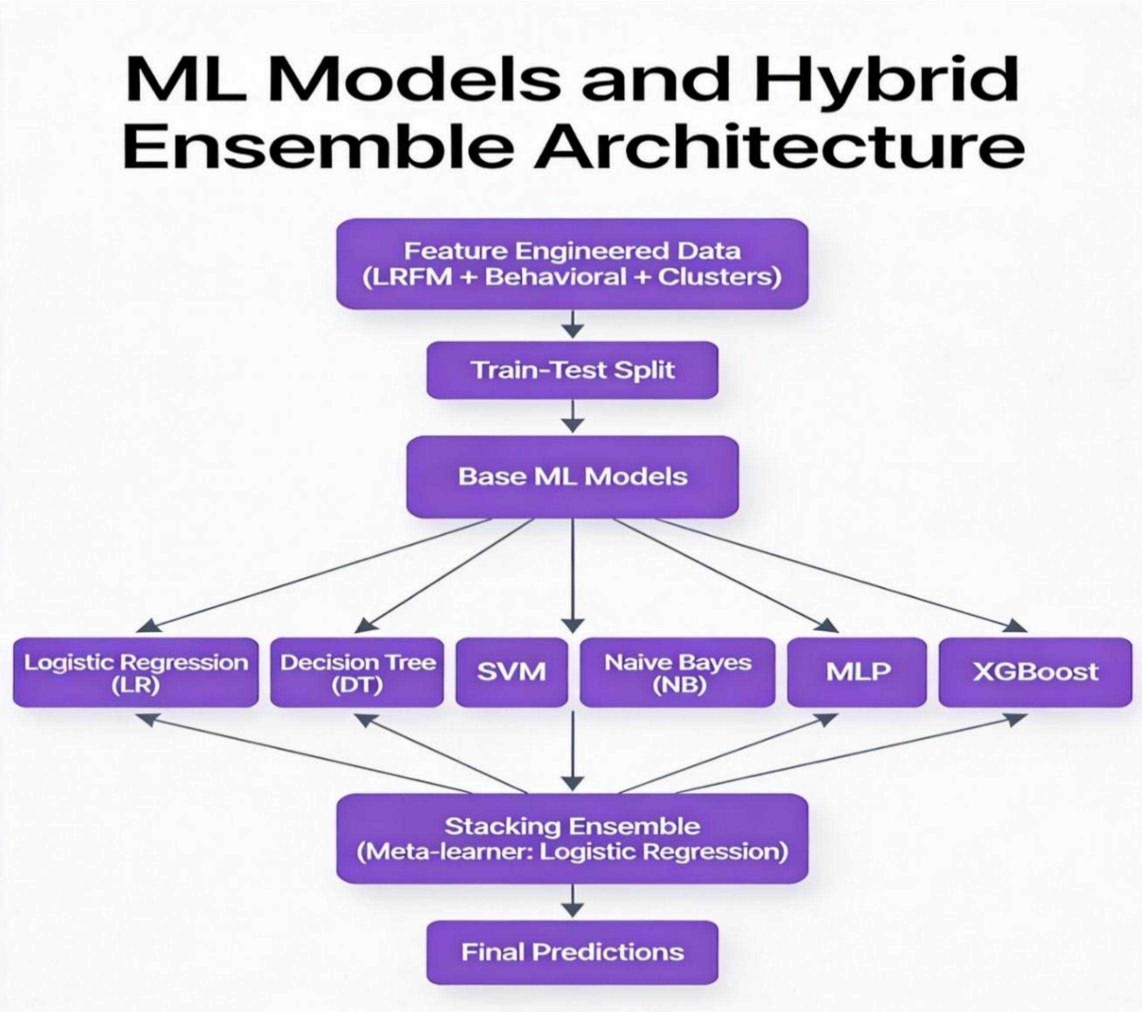


**Figure 5.6.2:** Level 1 DFD – Detailed Churn Prediction Workflow

**Explanation:** The Level 1 DFD extends the Level 0 diagram and presents the internal processing stages of the churn prediction system. It shows how data moves inside the system.

**Detailed Data Flow**

- Data Ingestion Module loads customer data from the October, November, and Telco datasets.
- Data Cleaning and Preprocessing Module removes missing values, fixes inconsistencies, and merges datasets into a unified format.
- LRFM Feature Extraction Module computes Length, Recency, Frequency, and Monetary values for each customer.

- Behavioural Feature Engineering Module generates indicators such as activity-drop ratio, normalized usage, spending trends, and recency weighting. Behaviour changes matter.
- Churn Label Generation Module assigns customers to Non-Churn, Partial Churn, or Total Churn categories based on observed patterns.
- Customer Segmentation Module applies K-Means clustering to group customers with similar behaviour.
- Train–Test Split and Feature Scaling Module prepares the dataset for model training and evaluation.
- Machine Learning Model Training Module trains multiple classifiers, including Logistic Regression, Random Forest, SVM, MLP, and XGBoost.
- Hybrid Stacking Ensemble Module combines predictions from all trained models to improve result stability.
- Model Evaluation Module measures performance using accuracy, precision, recall, F1-score, confusion matrix, and ROC analysis.
- Final Output Module generates the final churn classification for each customer. That is the output.

**Summary:**

The Level 1 DFD illustrates how raw customer data flows through preprocessing, feature extraction, modelling, and evaluation stages to produce actionable churn predictions.

# CHAPTER 6

# IMPLEMENTATION

## OVERVIEW

This chapter explains the implementation of the Multi-Stage Customer Churn Prediction System. It describes how raw customer data is converted into churn predictions using LRFM analysis, behavioural feature engineering, clustering, machine-learning models, and a hybrid stacking ensemble. The process is systematic. And practical.

## 6.1 ENVIRONMENT AND SETUP

The system is implemented using **Python (version 3.10)** because it provides strong support for data analysis and machine-learning tasks. It works well.

**Libraries Used:**

- pandas, numpy – handling datasets and numerical computations
- scikit-learn – machine-learning models, clustering, and preprocessing
- xgboost – gradient boosting classification
- matplotlib, seaborn – data visualization and performance analysis
- joblib – saving and loading trained models

This setup supports modular development and ensures consistent, repeatable execution.

## 6.2 DATASET LOADING AND CLEANING

**Purpose:** To ensure that only clean and reliable data is used for churn prediction. Data quality matters.

**Implementation Steps**

- Load October and November e-commerce datasets
- Load the Telco churn dataset for benchmarking
- Remove records with missing or invalid values
- Store cleaned datasets for further processing

**Explanation:** Data cleaning removes noise and inconsistencies that can affect feature accuracy and model results. Clean input leads to stable output.

## 6.3 LRFM FEATURE EXTRACTION ALGORITHM

**Purpose:** To represent long-term customer behaviour using four core metrics. Simple but effective.

**LRFM Metrics**

- **Length (L):** Duration of the customer's association with the platform
- **Recency (R):** Number of days since the last recorded activity
- **Frequency (F):** Count of interactions or purchases
- **Monetary (M):** Total spending value

**Implementation Steps**

- Group customer data using user_id
- Calculate frequency from interaction or purchase counts
- Compute recency using the most recent activity timestamp
- Aggregate monetary values per customer
- Measure overall engagement length

**Explanation:** LRFM condenses large transactional datasets into compact and meaningful customer profiles. Easier to model.

## 6.4 BEHAVIOURAL FEATURE ENGINEERING ALGORITHM

**Purpose:** To detect early behavioural decline (Partial Churn) that LRFM metrics alone may miss. Early signals matter.

**Features Implemented**

- Activity-drop ratio
- Normalized frequency
- Normalized monetary value
- Purchase-to-session ratio
- Recency weight
- Monetary trend

**Implementation Steps**

- Compare customer activity across October and November

- Compute decline ratios between months
- Normalize values to limit scale bias
- Apply recency-based weighting
- Handle zero and extreme values safely

**Explanation:** Behavioural features focus on change rather than volume, enabling early churn identification.

## 6.5 CHURN LABEL GENERATION ALGORITHM

**Purpose:** To convert observed behavioural changes into supervised learning labels.

**Churn Categories**

- **Non-Churn:** Stable or improved engagement
- **Partial Churn:** Reduced but continued activity
- **Total Churn:** Complete inactivity

**Implementation Steps**

- Compare activity levels across consecutive months
- Apply predefined threshold rules
- Assign churn labels accordingly

**Explanation**: This multi-class labelling approach reflects real customer behaviour more accurately than binary churn models.

## 6.6 TRAIN–TEST SPLIT AND FEATURE SCALING

**Purpose:** To ensure fair and unbiased model evaluation.

**Implementation Steps**

- Encode churn labels into numerical format
- Scale numerical features for consistency
- Apply stratified train–test split (80% training, 20% testing)

## 6.7 MACHINE LEARNING MODEL IMPLEMENTATION

**Purpose:** To learn churn patterns using different algorithmic approaches.

**Models Implemented**

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine,
- Naive Bayes,
- Multilayer Perceptron (MLP)
- XGBoost

**Explanation**:

Each model captures unique decision patterns, enabling effective performance comparison.

## 6.8 HYBRID STACKING ENSEMBLE ALGORITHM

**Purpose:** To improve prediction accuracy and stability by combining multiple models.

**Implementation Steps**

- Collect probability outputs from base models
- Stack probabilities into a single feature set
- Train Logistic Regression as the meta-learner
- Generate final churn predictions

**Explanation:** Stacking reduces single-model bias and serves as the core innovation of the system.

## 6.9 MODEL SAVING AND REUSABILITY

**Purpose:** To avoid repeated training and support efficient reuse.

**Implementation Steps**

- Save trained models using joblib
- Load models during evaluation or future runs

**Explanation:** This approach improves execution efficiency and supports deployment readiness.

# CHAPTER 7

# SYSTEM EVALUATION AND TESTING

## OVERVIEW

This chapter evaluates the performance, correctness, and reliability of the proposed Multi-Stage Customer Churn Prediction System. The evaluation checks whether the system correctly classifies customers into Non-Churn, Partial Churn, and Total Churn categories. Testing is carried out at multiple levels. It is thorough.The evaluation covers dataset quality, feature correctness, model behaviour, and ensemble reliability.

## 7.1 EVALUATION STRATEGY

The evaluation strategy focuses on both prediction accuracy and system correctness. Since the system is machine-learning based, testing relies on quantitative metrics and visual analysis rather than simple pass or fail checks.

Evaluation is carried out in three stages:saq

- Feature-level validation
- Model-level performance evaluation
- System-level testing and comparison

## 7.2 DATASET VALIDATION AND PREPROCESSING TESTING

**Purpose:** To confirm that input datasets are clean, consistent, and suitable for churn modelling.

**Testing Performed**

- Verified successful loading of October and November datasets
- Confirmed removal of missing and inconsistent values
- Validated correct dataset merging using user_id
- Ensured LRFM and behavioural features contained no null or infinite values

**Outcome:** All datasets were processed correctly. They were suitable for multi-class churn prediction.

## 7.3 FEATURE VALIDATION TESTING

**Purpose:** To verify the correctness of computed LRFM and behavioural features.

**Tests Conducted**

- Analyzed recency distributions to check realistic time gaps
- Examined frequency and monetary distributions for skewness and outliers
- Validated activity-drop ratios across consecutive months

**Outcome**: The features showed meaningful patterns. They accurately represented both long-term and short-term customer behaviour.

## 7.4 MODEL PERFORMANCE EVALUATION

**Purpose:** To assess the predictive ability of individual machine learning models.

**Models Evaluated**

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine
- Naive Bayes
- Multilayer Perceptron (MLP)
- XGBoost

**Evaluation Metrics**

- Accuracy
- Precision
- Recall
- F1-Score

**Results:** Most models achieved strong accuracy. Tree-based and ensemble models performed better than simpler classifiers. This confirms the usefulness of LRFM and behavioural features.

## 7.5 HYBRID STACKING ENSEMBLE EVALUATION

**Purpose:** To assess the effectiveness of the hybrid ensemble approach.

**Evaluation Approach:**

- Combined probability outputs from base models
- Used Logistic Regression as a meta-learner

- Compared ensemble performance with individual models

**Outcome:**

The hybrid ensemble produced more stable predictions. It reduced Partial Churn misclassification and showed better generalization than single models.

## 7.6 CONFUSION MATRIX ANALYSIS

**Purpose:** To study prediction behaviour for each churn category.

**Testing Conducted:**

- Generated confusion matrices for all churn classes
- Examined confusion between Partial Churn and Total Churn

**Outcome:** Strong diagonal values were observed. This indicates accurate class-wise predictions and clear separation between churn categories.

## 7.7 ROC CURVE AND CLASSIFICATION REPORT ANALYSIS

**Purpose:** To evaluate class separation and prediction confidence.

**Testing Conducted:**

- Generated ROC curves for multi-class churn prediction
- Analyzed class-wise precision, recall, and F1-scores

**Outcome:** High ROC values and balanced precision–recall scores indicate reliable performance and strong class discrimination.

## 7.8 SYSTEM TESTING

### 7.8.1 Functional Testing

The following checks were performed:

- Successful data loading
- Correct feature generation
- Proper model training and prediction
- Accurate saving and loading of trained models

All functional components worked as expected.

### 7.8.2 Reliability Testing

The system was executed multiple times using the same dataset

- Predictions remained consistent across runs

This confirms system stability. And reproducibility.

### 7.8.3 Scalability Testing

System performance was tested with larger datasets

- Feature computation and model execution remained efficient

The system scaled well.

## 7.9 COMPARATIVE ANALYSIS WITH EXISTING SYSTEMS

| Aspect | Existing Systems | Proposed System |
|---|---|---|
| Churn Type | Binary | Multi-Class (Non, Partial, Total) |
| Feature Depth | Basic transactional | LRFM + Behavioural |
| Early Detection | Not supported | Supported (Partial Churn) |
| Model Strategy | Single model | Hybrid stacking ensemble |
| Interpretability | Limited | Enhanced via clustering |

The proposed system overcomes the key limitations of traditional churn prediction approaches.

## SUMMARY OF EVALUATION

The evaluation and testing phase confirms that:

- Customer data is handled correctly throughout the pipeline.
- LRFM and behavioural features are generated with accuracy.
- Machine-learning models deliver consistent and reliable results.
- The hybrid ensemble improves prediction robustness.
- Partial churn is identified effectively. Early action becomes possible.

# CHAPTER 8

# RESULT ANALYSIS

## OVERVIEW

This section examines the results produced during the implementation and evaluation of the Multi-Stage Customer Churn Prediction System. It reviews observed customer behaviour patterns, the role of LRFM and behavioural features, the performance of individual machine-learning models, and the impact of the hybrid stacking ensemble. Each aspect is assessed carefully. The results show that the proposed system accurately classifies customers into Non-Churn, Partial Churn, and Total Churn categories. It also overcomes key limitations found in traditional churn prediction methods.

## 8.1 DISCUSSION ON DATA UNDERSTANDING RESULTS

The data understanding phase revealed noticeable differences in customer behaviour. Recency analysis showed that active users interacted recently, while a smaller group displayed long inactivity periods, indicating higher churn risk. Frequency and Monetary distributions were skewed, where a small number of users contributed high engagement and spending, while many showed low or declining activity. Churn is not uniform. It varies. These findings highlight the need to analyze both long-term behaviour and recent activity changes.

## 8.2 IMPACT OF LRFM FEATURES

LRFM features played an important role in distinguishing customer value levels:

- **Length** identified loyal and long-term users
- **Recency** separated active customers from inactive ones
- **Frequency** reflected the intensity of platform usage
- **Monetary** represented customer revenue contribution

However, LRFM features alone could not clearly identify Partial Churn. Some users remained active but with reduced intensity. This gap required additional behavioural indicators.

## 8.3 CONTRIBUTION OF BEHAVIOURAL FEATURES

Behavioural features improved churn detection significantly:

- **Activity-drop ratio** captured early engagement decline

- **Normalized frequency and monetary values** reduced bias from high-activity users
- **Recency weighting** emphasized recent behaviour patterns
- **Monetary trend** highlighted early reductions in spending

These features enabled reliable identification of Partial Churn. Traditional models often miss it.

## 8.4 DISCUSSION ON CUSTOMER SEGMENTATION RESULTS

K-Means clustering grouped customers based on similar behaviour patterns. The resulting clusters identified:

- Highly active and loyal users
- Moderately active users with declining engagement
- Low-activity or disengaged users

This segmentation improved interpretability. Risk patterns became clearer.

## 8.5 PERFORMANCE OF INDIVIDUAL MACHINE LEARNING MODELS

The evaluation of individual models showed the following:

- Tree-based models (Decision Tree, Random Forest, XGBoost) performed strongly due to their ability to handle non-linear behaviour.
- Logistic Regression offered a stable and interpretable baseline.
- SVM and Naive Bayes showed comparatively lower performance because they are sensitive to feature distribution.
- MLP learned complex patterns but required proper feature scaling for consistent results.

No single model performed best in all cases. This confirmed the need for ensemble learning.

## 8.6 EFFECTIVENESS OF THE HYBRID STACKING ENSEMBLE

The hybrid stacking ensemble showed the most consistent results across models:

- It reduced misclassification when compared to individual classifiers
- It maintained stable performance across all churn categories
- It performed strongly in identifying Partial Churn cases

By combining different learning approaches, the ensemble confirmed the effectiveness of the chosen system architecture.

## 8.7 CONFUSION MATRIX AND CLASS-WISE ANALYSIS

The confusion matrix displayed clear diagonal dominance, showing that most customers were classified correctly. Very little overlap occurred between Partial Churn and Total Churn classes. This confirms the system's ability to separate early behavioural decline from full disengagement.

Class-wise accuracy results remained balanced. That matters.

## 8.8 COMPARISON WITH TRADITIONAL CHURN PREDICTION SYSTEMS

Traditional churn prediction systems typically:

- Use binary churn classification
- Depend on single models
- Miss gradual behaviour changes

In contrast, the proposed system:

- Supports multi-class churn prediction
- Detects early warning signs using behavioural features
- Applies hybrid ensemble learning for improved reliability
- Enhances interpretability through customer clustering

This highlights the practical value of the proposed approach

## 8.9 PRACTICAL IMPLICATIONS OF RESULTS

The results show that organizations can:

- Identify at-risk customers earlier
- Plan focused retention actions
- Reduce potential revenue loss
- Improve customer lifetime value

Early detection of Partial Churn allows timely action. Before customers leav

## SUMMARY OF RESULT DISCUSSION

The result analysis confirms that:

- LRFM and behavioural features work effectively together
- Multi-class churn prediction provides clearer insights
- Hybrid ensemble learning improves stability and accuracy

Overall, the proposed system performs better than traditional churn models in detection capability, reliability, and interpretability.

# CHAPTER 9

# SNAPSHOTS

## 9.1 DATA UNDERSTANDING GRAPHS

### 9.1.1 Recency Distribution

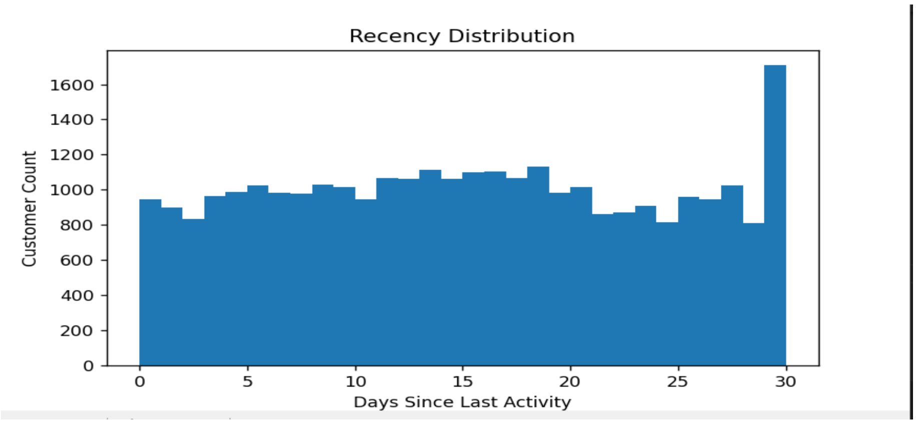


**Figure 9.1.1:** Recency Distribution

**Explanation:** This graph presents customer distribution based on the number of days since the last recorded activity. Lower recency values indicate active users, while higher values point to inactivity. The spread clearly highlights customers at churn risk, particularly those in Partial and Total Churn stages.

### 9.1.2 Frequency Distribution

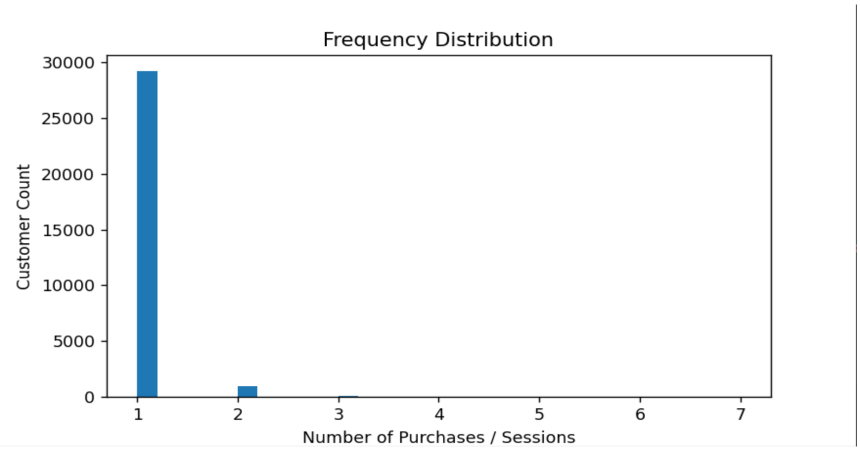


**Figure 9.1.2:** Frequency Distribution

**Explanation:** The frequency distribution shows how often customers interact or make purchases. Most users appear with low activity levels, while a small segment shows frequent engagement. This contrast helps separate loyal customers from those slowly disengaging.

## 9.2 CHURN ANALYSIS GRAPH

### 9.2.1 Churn Distribution

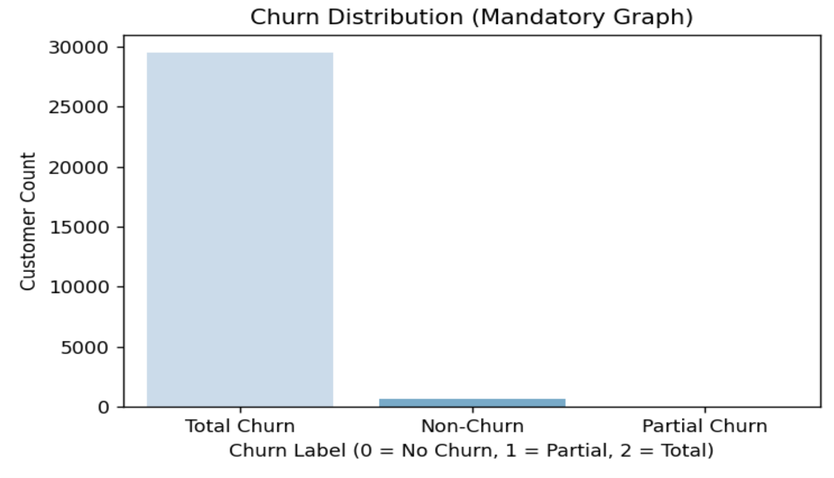


**Figure 9.2.1:** Churn Distribution

**Explanation:** This graph displays the count of customers classified as Non-Churn, Partial Churn, and Total Churn. The presence of Partial Churn supports early churn detection, while the higher Total Churn count reflects the severity of customer drop-off.

## 9.3 MODEL PERFORMANCE EVALUATION

### 9.3.1 Model Accuracy Comparison

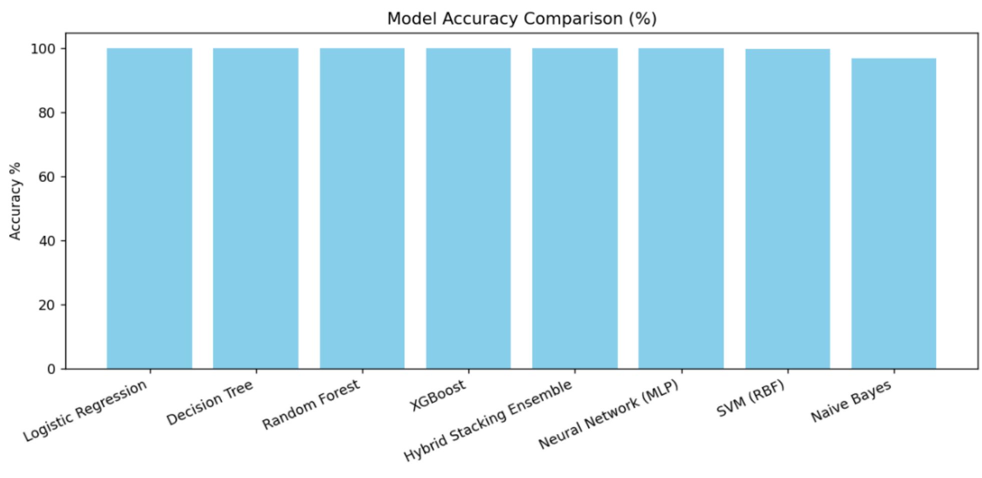

**Figure 9.3.1:** Model Accuracy Comparison

**Explanation:** This figure compares prediction accuracy across different machine-learning models. Tree-based and ensemble models achieve better results, supporting the use of a hybrid stacking ensemble for more stable churn prediction.

### 9.3.3 Confusion Matrix

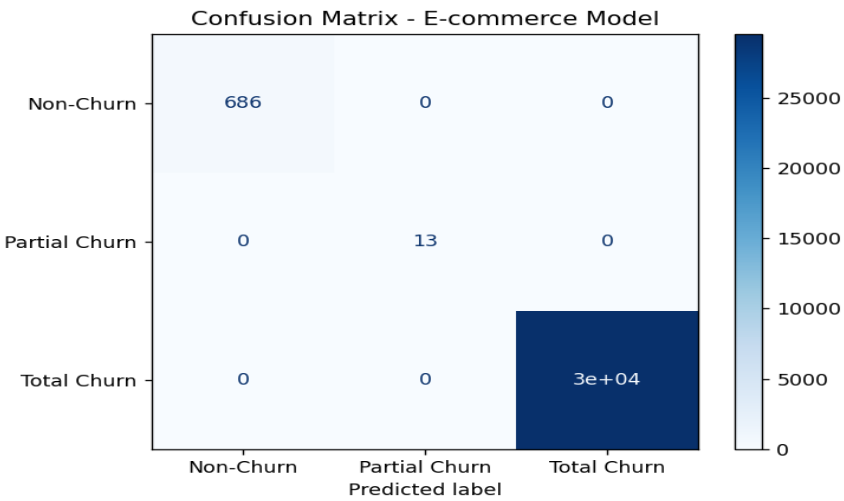


**Figure 9.3.3:** Confusion Matrix – Churn Classification

**Explanation:** The confusion matrix summarizes correct and incorrect predictions for each churn class. Strong diagonal values indicate accurate classification, with very little confusion between Partial and Total Churn categories.

### 9.3.4 ROC Curve -Multi Class XGBoost Model

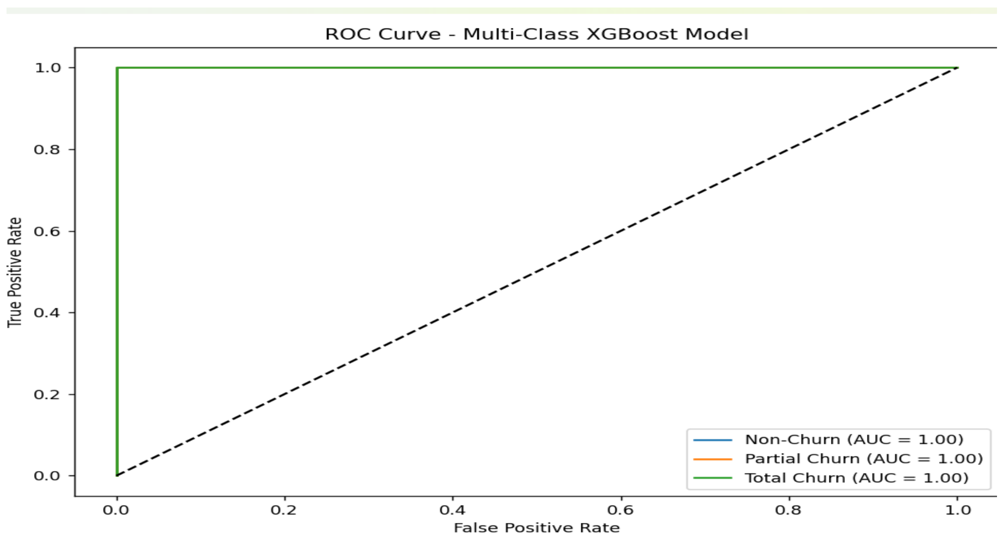

**Figure 9.3.4**: ROC Curve – Multi-Class XGBoost Model

**Explanation:** The ROC curve shows the relationship between true positive and false positive rates for each churn class. High AUC values confirm effective separation among Non-Churn, Partial Churn, and Total Churn customers.

## 9.4 CHURN TYPE EVALUATION

### 9.4.1 Class-wise Accuracy for Churn Types

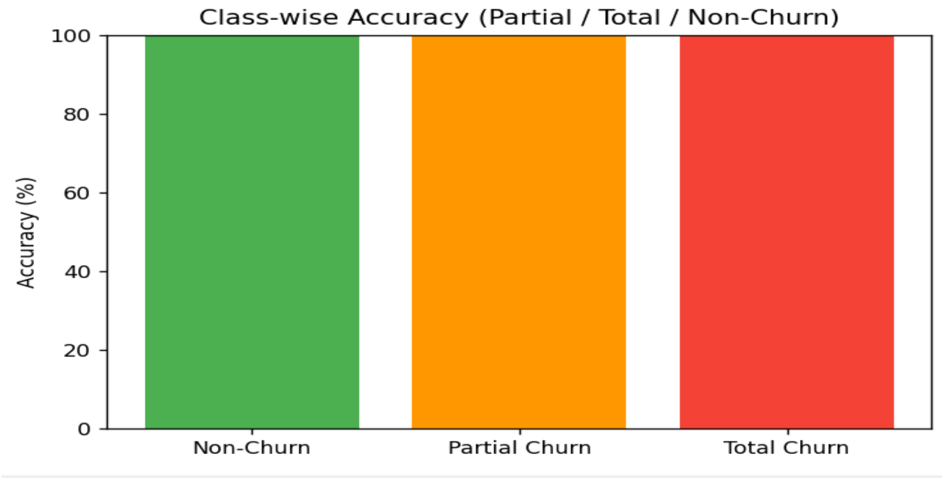


**Figure 9.4.1:** Class-wise Accuracy for Churn Types

**Explanation:** The ROC curve shows the balance between true positive and false positive rates for each churn class. High AUC values indicate clear separation between Non-Churn, Partial Churn, and Total Churn categories. The distinction is strong.

## 9.5 MODEL INSIGHTS & CROSS-DOMAIN ANALYSIS

### 9.5.1 Feature Importance – XGBoost Model

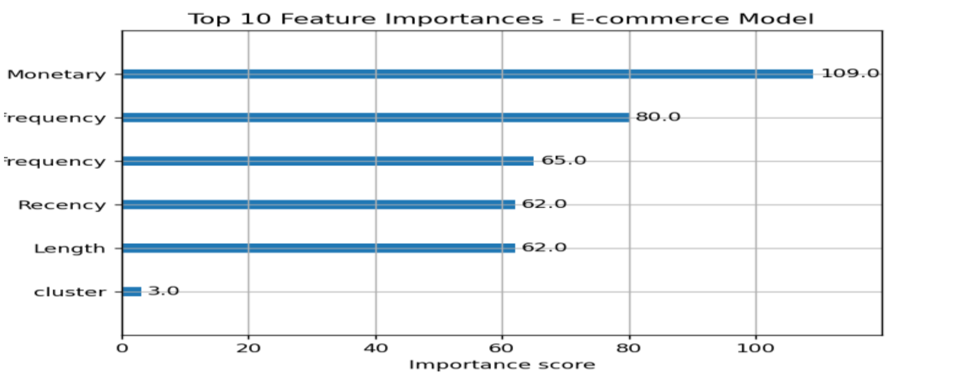


**Figure 9.5.1:** Feature Importance – XGBoost Model

**Explanation:** This graph highlights the key features influencing churn prediction. Recency, frequency, and monetary attributes carry the highest weight, showing that customer activity and spending patterns directly affect churn behaviour. Usage matters

### 9.5.2 Ecommerce vs Telco Model Accuracy Comparison

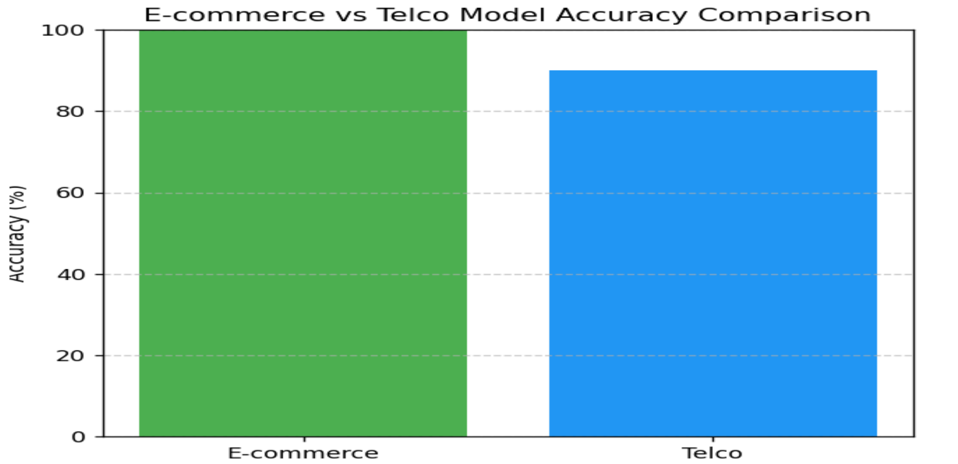


**Figure 9.5.2:** E-commerce vs Telco Accuracy Comparison

**Explanation:** This comparison evaluates model performance on both e-commerce and telco datasets. Similar accuracy levels across domains confirm that the system adapts well beyond a single dataset. It scales reliably.

## 9.6 TERMINAL OUTPUT SNAPSHOTS

**November dataset splits (nov_dataset_split)**
**File: train.csv**
  Rows: 24938
  Counts:
    Non-Churn: 552
    Partial Churn: 10
**File: test.csv**
  Rows: 6235
  Counts:
    Non-Churn: 156
    Partial Churn: 3

**October dataset splits (oct_dataset_split)**
**File: train.csv**
  Rows: 25072
  Counts:

Total Churn: 23786
Non-Churn: 548
Partial Churn: 13

**File: test.csv**
Rows: 6269
Counts:
Total Churn: 6050
Non-Churn: 159
Partial Churn: 6

**Telco dataset splits (telco_dataset_split)**

**File: train.csv**
Rows: 5000
Counts:
No: 3687
Yes: 1313

**File: test.csv**
Rows: 2000
Counts:
No: 1456
Yes: 544

**OVERALL (merged_LRFM_labeled.csv) churn distribution**

**churn_label**

| | |
|---|---|
| Total Churn | 29519 |
| Non-Churn | 686 |
| Partial Churn | 13 |

**Figure 9.12:** Console Output of Train–Test Dataset Splits and Churn Class Distribution

**Explanation:** This snapshot displays the train–test split and the distribution of churn classes used for model training. It verifies correct dataset preparation and balanced class representation for supervised learning. The setup is valid.

## SUMMARY OF SNAPSHOTS

This chapter presents selected graphical outputs and terminal results generated during system execution. These snapshots support data understanding, churn analysis, model performance evaluation, and cross-domain validation.

# CHAPTER 10

# CONCLUSION

Customer churn prediction remains a major challenge for digital platforms because retaining existing users costs far less than acquiring new ones. Most traditional churn models treat churn as a binary outcome and focus only on users who completely stop using a service. This approach misses early behaviour changes that often appear before full disengagement.

This project addresses the gap by designing and implementing a multi-stage churn prediction framework that identifies Non-Churn, Partial Churn, and Total Churn customers. The system combines LRFM-based long-term behaviour analysis with short-term behavioural features to understand customer activity more clearly.

Indicators such as activity-drop ratio, recency weighting, and monetary trends helped identify users whose engagement was declining but had not yet stopped. Early signals matter. This enabled timely detection of Partial Churn and supported proactive retention actions.

The system trained and evaluated multiple machine-learning models to capture different learning patterns and decision boundaries. A Hybrid Stacking Ensemble then combined individual model outputs, improving prediction stability and consistency. Evaluation results showed strong performance across all churn categories, with high accuracy, balanced precision and recall, clear confusion matrix separation, and reliable ROC results.

The system was further tested on datasets from both e-commerce and telecom domains. Results remained consistent. This confirms adaptability. Overall, the project demonstrates that multi-class churn prediction supported by behavioural analytics and ensemble learning provides a practical and interpretable solution for churn analysis. The proposed framework offers a solid base for real-world customer retention systems.

# FUTURE ENHANCEMENTS

While the proposed system produced stable and dependable results, several improvements can further enhance its usability and scope:

- **Temporal and Sequential Modelling:**
  Future work can model customer behaviour across time using sequence-based methods such as LSTM or GRU networks. This can improve early churn detection. Especially for gradual decline.
- **Real-Time Churn Risk Scoring:**
  The system can be extended to process live customer activity streams, enabling instant churn risk alerts and faster business response.
- **Explainable Artificial Intelligence (XAI):**
  Techniques like SHAP or LIME can be integrated to explain model predictions, helping stakeholders understand churn drivers. Transparency improves trust.
- **Automated Retention Recommendation Engine**:
  A recommendation module can suggest targeted offers or engagement actions based on predicted churn category and customer segment.
- **Scalability and Cloud Deployment:**
  Deploying the system on cloud platforms with distributed frameworks such as Apache Spark can support large datasets and high user traffic.
- **Advanced Hyperparameter Optimization:**
  Methods like Bayesian Optimization or AutoML can refine model parameters and improve prediction accuracy.
- **Cross-Domain and Cross-Platform Expansion:**
  The framework can be adapted for domains such as banking, subscription services, and SaaS with minimal changes.
- **Integration with Business Dashboards:**
  Interactive dashboards can display churn trends, risk scores, and customer segments to support data-driven decision-making.

# REFERENCES


1. Your Novel Paper, “A Novel Model for Partial and Total Churn Prediction,” 2024.
2. T. Fawcett and F. Provost, “Combining Data Mining and Machine Learning for Effective Customer Retention,” KDD, 1997.
3. J. Hadden et al., “Computer Assisted Customer Churn Management,” Computers & Operations Research, 2007.
4. A. Ahmed and D. Maheswari, “Customer Churn Prediction using Machine Learning,” IJARCS, 2018.
5. C. Verbeke et al., “Predictive Modelling for Churn Using Ensemble Methods,” Expert Systems with Applications, 2012.
6. Y. Freund and R. Schapire, “A Decision-Theoretic Generalization of Online Learning,” JCSS, 1997.
7. T. Chen and C. Guestrin, “XGBoost: A Scalable Tree Boosting System,” *KDD*, 2016.
8. J. Han, M. Kamber, and J. Pei, Data Mining: Concepts and Techniques, Morgan Kaufmann, 2012.
9. I. Goodfellow et al., Deep Learning, MIT Press, 2016.
10. S. Maklan and P. Klaus, “Customer Experience and Churn,” Journal of Marketing Management, 2011.
11. S. Bose and X. Chen, “Hybrid Machine Learning for Churn Prediction,” IEEE Access, 2019.
12. A. Ng, “Machine Learning and AI via Brain Simulations,” Neural Information Processing, 2017.
13. F. Reichheld, “The Loyalty Effect,” Harvard Business School Press, 1996.
14. J. Quinlan, “Induction of Decision Trees,” Machine Learning, 1986.
15. L. Breiman, “Random Forests,” Machine Learning, 2001.
16. T. Hastie, R. Tibshirani, and J. Friedman, The Elements of Statistical Learning, Springer, 2009.

17. C. Bishop, Pattern Recognition and Machine Learning, Springer, 2006.

18. K. Coussement and D. Van den Poel, "Churn Prediction in Online Services," Decision Support Systems, 2008.

19. A. Geron, Hands-On Machine Learning, O'Reilly, 2019.

20. S. Aggarwal, Neural Networks and Deep Learning, Springer, 2018.

21. M. Kuhn and K. Johnson, Applied Predictive Modelling, Springer, 2013.

22. J. Brownlee, Machine Learning Mastery, 2020.

23. D. Hand, "Classifier Technology and the Illusion of Progress," Statistical Science, 2006.

24. P. Domingos, "A Few Useful Things to Know About Machine Learning," *CACM*, 2012.

25. S. Burez and D. Van den Poel, "CRM and Churn Prediction," Expert Systems, 2009.

26. R. Sharma et al., "Partial and Total Churn Prediction Using Behavioural Analytics," IEEE Access, 2024.

27. F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *JMLR*, 2011.

28. K. Murphy, Machine Learning: A Probabilistic Perspective, MIT Press, 2012.

29. D. Powers, "Evaluation: Precision, Recall and F-Measure," Journal of Machine Learning, 2011.

30. S. Russell and P. Norvig, Artificial Intelligence: A Modern Approach, Pearson, 2020.